

LECTURE-25

wednesday
6/5/20.

Q:

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

lowest in grid on AB

d	K	Clusters
0	4	A, B, C, D
1	3	{A, B}, C, D
3	2	{A, B}, {C, D}
6	1	{A, B}, {C, D}

	AB	C	D
AB	0	4	6
C		0	3
D			0

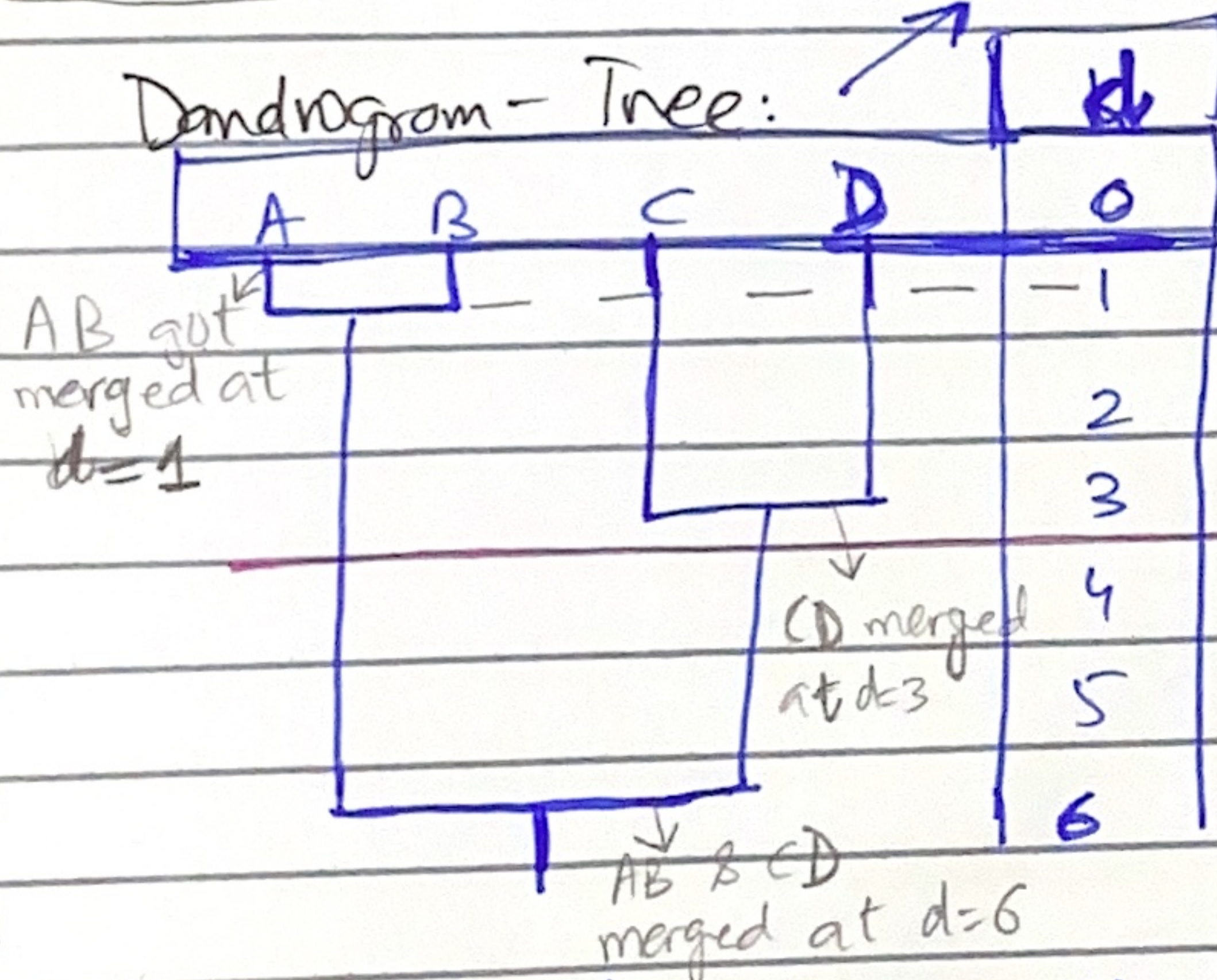
$$\begin{matrix} A \rightarrow C = 4 \\ B \rightarrow C = 2 \end{matrix} \} \text{Max} = 4.$$

lowest in grid at CD

	AB	CD
AB	0	6
CD		0

Used to see which clusters are most stable
 $\therefore$ hamain K may clusters (value of K) rakha hai for optimum sol.

Dendrogram - Tree:



* From 'dKC' table:

We'll find stability of each value of 'K' by seeing how long did it survive & that will be done via 'd' column which basically is treated as time.

$$K=4 \rightarrow d=0, 1 \rightarrow \text{Stayed for } 1-0 = \boxed{1} \text{ unit}$$

$$K=3 \rightarrow d=1, 3 \rightarrow \text{Stayed for } 3-1 = \boxed{2}$$

$$K=2 \rightarrow d=3, 6 \rightarrow \text{Stayed for } 6-3 = \boxed{3}$$

Answer: Since 'K=2' stayed for longest, means it is most stable and we'll choose this as optimal solution.

- Put a line on dendrogram just below '3' because at $K=2$; $d=3$.
- The no. of optimal clusters is '2' as $K=2$.

Note: • Min. func. in clustering makes elongated ^{clusters} and they're closer to each other

- Max. func. makes spherical and far far clusters.
- Avg. is mixture of both characteristics.